

1. Review

Differential Equations

↳ Ordinary differential equation

↳ a) Order b) Degree c) Linearity

d) Homogeneous/Non-homogeneous

e) Solution of differential equation

f) Interval of existence

g) Explicit and Implicit solution

h) General and particular solution

i) Initial condition

↳ First Order Ordinary Differential Equations

↳ Separable First Order ODEs.
General and Particular solutionsToday's topic :- Homogeneous First Order ODEs

Review: 2

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Seperable First Order ODEs

In general $\frac{dy}{dx} = F(x, y) \leadsto \textcircled{1}$

if we can write

$$\frac{dy}{dx} = f(x)g(y) \leadsto \textcircled{2}$$

then the differential equation seperable

Method of solution.

step I:- Seperate Independent and dependent variables

i.e. $\frac{dy}{g(y)} = f(x) dx \quad (\text{from } 2)$

step II:- Integrate both sides with their respective variables.

i.e. $\int \frac{1}{g(y)} dy = \int f(x) dx$

This gives general solution in explicit or implicit form.

Example ①

③

$$(1-x)dy - (1+y)dx = 0$$

$$(1-x)dy = (1+y)dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{1+y}{1-x}$$

or $\boxed{\frac{dy}{1+y} = \frac{dx}{1-x}}$ Separable form

Integrate both sides

$$\int \frac{dy}{1+y} = \int \frac{dx}{1-x}$$

$$\ln(1+y) = -\ln(1-x) + \ln C$$

$$\ln(1+y) + \ln(1-x) = \ln C$$

$$\ln[(1+y)(1-x)] = \ln C$$

$$(1+y)(1-x) = C$$

General Solution in Implicit form.

Example ②

④

$$(1-x^2)y' + y = 0 \quad y(1) = 2$$

$$\Rightarrow y' = \frac{dy}{dx} = -(1-x^2)y$$

$$\frac{dy}{y} = -(1-x^2)dx$$

$$\int \frac{1}{y} dy = - \int (1-x^2) dx$$

$$-\frac{1}{y} = -x + \frac{x^3}{3} + C$$

$$\text{or } \boxed{\frac{1}{y} = x - \frac{x^3}{3} + D} \quad (-C = D)$$

General solution.

Now using initial condition

$y(1) = 2$; that is $x=1; y=2$
use in general solution

$$\frac{1}{2} = 1 - \frac{1}{3} + D$$

$$D = \frac{1}{2} - \frac{2}{3} = -\frac{1}{6} \quad \text{use in general solution}$$

$$\boxed{\frac{1}{y} = x - \frac{x^3}{3} - \frac{1}{6}} \quad \text{Particular Solution.}$$

10. Show that the problem $y' = 2x$, $y(0) = 0$, $y(1) = 100$, has no solution. Is this an initial value problem?

1.2 SEPARABLE EQUATIONS

A first-order differential equation that can be written in the form

$$p(y) \frac{dy}{dx} = q(x), \quad (1.7)$$

where p and q are given functions, is called a *separable* equation. Examples of such equations are

$$y^{-2} \frac{dy}{dx} = 2x, \quad y^{-1} \frac{dy}{dx} = (x+1)^{-1}, \quad (3y^2 + e^y) \frac{dy}{dx} = \cos x.$$

If a function f is a solution of Eq. (1.7) on an interval $\mathcal{I}$, then

$$p[f(x)]f'(x) = q(x)$$

for x in $\mathcal{I}$. Taking antiderivatives, we have

$$\int p[f(x)]f'(x) dx = \int q(x) dx + c$$

or

$$\int p(y) dy = \int q(x) dx + c.$$

If P and Q are functions such that $P'(y) = p(y)$ and $Q'(x) = q(x)$, then the solution f must satisfy the equation

$$P(y) = Q(x) + c, \quad (1.8)$$

where c is a constant. That is,

$$P[f(x)] = Q(x) + c$$

for x in $\mathcal{I}$. Conversely, if y is any differentiable function that satisfies Eq. (1.8), we see by implicit differentiation that

$$P'(y) \frac{dy}{dx} = Q'(x)$$

or

$$p(y) \frac{dy}{dx} = q(x).$$

Thus, a function is a solution of Eq. (1.7) if and only if it satisfies an equation of the form (1.8) for some choice of the constant c . It may not be possible to find an explicit formula for y in terms of x from Eq. (1.8). However, we say that Eq. (1.8) determines the solutions of the differential equation *implicitly*. Let us now consider some examples of separable equations.

Example 1

$$\frac{dy}{dx} = 2xy^2. \quad (1.9)$$

“Separating the variables,” we have

$$y^{-2} \frac{dy}{dx} = 2x$$

or

$$y^{-2} dy = 2x dx.$$

Taking antiderivatives, we have

$$\int y^{-2} dy = \int 2x dx + c$$

or

$$-\frac{1}{y} = x^2 + c.$$

Thus, the functions defined by the formula

$$y = \frac{-1}{x^2 + c} \quad (1.10)$$

are solutions of Eq. (1.9). Note, however, that the identically zero function ($y = 0$) is also a solution. In arriving at formula (1.10), we started out by dividing both sides of the original equation by y^2 , and this procedure is not valid when $y = 0$.

Suppose that it is desired to find the solution curve that passes through the point $(2, -1)$ in the xy plane. Then our initial condition is

$$y(2) = -1.$$

Setting $x = 2$ and $y = -1$ in formula (1.10), we see that

$$-1 = \frac{-1}{4 + c}$$

or $c = -3$. Then the desired solution is given by the formula

$$y = \frac{1}{3 - x^2}.$$

Example 2

$$(x + 1) \frac{dy}{dx} = 2y.$$

Here we have

$$\frac{dy}{y} = 2 \frac{dx}{x + 1}$$

or

$$\ln |y| = \ln (x + 1)^2 + c'.$$

Then

$$|y| = e^{c'}(x + 1)^2$$

and

$$y = \pm e^{c'}(x + 1)^2, \quad (1.11)$$

where c' is an arbitrary constant. But $\pm e^{c'}$ can have any value except zero, so the set of functions described by formula (1.11) is also described by the simpler formula

$$y = c(x + 1)^2, \quad (1.12)$$

where c is a constant different from zero but otherwise is arbitrary. However, since $y = 0$ is obviously a solution of the differential equation, formula (1.12) also describes a solution when $c = 0$. This formula, with c completely arbitrary, gives the general solution of the equation.

Example 3 Consider the initial value problem

$$\frac{dy}{dx} = \frac{\cos x}{3y^2 + e^y}, \quad y(0) = 2.$$

From the differential equation we have

$$(3y^2 + e^y) dy = \cos x dx$$

or

$$y^3 + e^y = \sin x + c.$$

Setting $x = 0$ and $y = 2$ (these values come from the initial condition) we find that

$$8 + e^2 = c.$$

Homogeneous First Order ODEs.

(5)

In general

$$\frac{dy}{dx} = F(x, y)$$

If we can write

$$\frac{dy}{dx} = \frac{f(x, y)}{g(x, y)}$$

and $f(x, y)$ and $g(x, y)$ both are of the same degree,
then the differential equation is homogeneous

Note:- $f(x, y)$ and $g(x, y)$ are to homogeneous functions of x & y .

Examples Homogeneous Functions

$$f(x, y) = x^3 - 4x^2y + xy^2 - 10y^3 \quad \left[\begin{array}{l} \text{Homogeneous} \\ \text{Degree 3} \end{array} \right]$$

$$g(x, y) = x^3 - \frac{3x^4 + 5y^4}{y} \quad [\text{Homogeneous, Degree 3}]$$

$$h(x, y) = x^2 - y^2 \quad [\text{Homogeneous, Degree 2}]$$

$$k(x, y) = x^2 - 6xy - 4 \quad [\text{Non-homogeneous}]$$

$$p(x, y) = x^3 - x^2y - y^3 - 6xy \quad [\text{Non-homogeneous}]$$

Example and Method of solution

(6)

Example ①

$$(x^2 + y^2)dx + 2xy dy = 0$$

$$\left[\frac{dy}{dx} = \frac{-(x^2 + y^2)}{2xy} \right] = \frac{f(x, y)}{g(x, y)} \quad \left[\begin{array}{l} \text{Homogeneous} \\ \text{Degree 2} \end{array} \right]$$

①

Solution: Step I

Let $y = vx$
and $\frac{dy}{dx} = v + \frac{dv}{dx} x$ } use in ①

$$v + x \frac{dv}{dx} = \frac{-(x^2 + v^2 x^2)}{2x(vx)} = \frac{-(x^2 + v^2 x^2)}{2x^2 v} = \frac{-\cancel{x^2}(1 + v^2)}{2\cancel{x^2} v}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{-(1 + v^2)}{2v}$$

$$x \frac{dv}{dx} = \frac{-(1 + v^2)}{2v} - v = \frac{-1 - v^2 - 2v^2}{2v} = \frac{-(1 + 3v^2)}{2v}$$

$$\boxed{x \frac{dv}{dx} = \frac{-(1 + 3v^2)}{2v}}$$

Seperable first order ODE.

$$\frac{-2v}{1 + 3v^2} dv = \frac{1}{x} dx$$

Integrate both sides.

(7)

$$\int \frac{-2v}{1+3v^2} dv = \int \frac{1}{x} dx$$

$$-\frac{1}{3} \int \frac{6v}{1+3v^2} dv = \int \frac{1}{x} dx$$

$$-\frac{1}{3} \ln(1+3v^2) = \ln x + \ln C$$

$$3 \ln C = \ln D$$

$$-\ln(1+3v^2) = 3 \ln x + \ln D$$

$$-\ln D = \ln E$$

$$3 \ln x + \ln(1+3v^2) = \ln E$$

$$\ln x^3 + \ln(1+3v^2) = \ln E$$

$$\ln(x^3(1+3v^2)) = \ln E$$

$$\boxed{x^3(1+3v^2) = E}$$

Backward Substitution

$$y = vx \Rightarrow v = \frac{y}{x}$$

$$x^3 \left(1 + 3 \frac{y^2}{x^2} \right) = E$$

$$x^3 \left(\frac{x^2 + 3y^2}{x^2} \right) = E$$

$$\boxed{x(x^2 + 3y^2) = E}$$

Ans.

Hence, the desired solution (if such a solution exists) is implicitly determined by the equation

$$y^3 + e^y = \sin x + 8 + e^2.$$

Some differential equations that are not separable as they stand become separable after a change of variable. One such class of equations consists of those that can be written in the form

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right). \quad (1.13)$$

An example of such an equation is

$$(x^4 + y^4) \frac{dy}{dx} = x^3 y,$$

which may be rewritten as

$$\frac{dy}{dx} = \frac{x^3 y}{x^4 + y^4}$$

or

$$\frac{dy}{dx} = \frac{y/x}{1 + (y/x)^4}.$$

An equation of the form (1.13) can be made separable by introducing a new independent variable v where

$$v = \frac{y}{x}.$$

For then

$$y = vx, \quad \frac{dy}{dx} = x \frac{dv}{dx} + v,$$

and Eq. (1.13) becomes

$$x \frac{dv}{dx} + v = F(v)$$

or

$$\frac{1}{F(v) - v} \frac{dv}{dx} = \frac{1}{x},$$

which is separable.

As an example we consider the equation

$$2x^2 \frac{dy}{dx} = x^2 + y^2.$$

This equation can be put in the form (1.21) since

$$\frac{dy}{dx} = \frac{x^2 + y^2}{2x^2} = \frac{1}{2} \left[1 + \left(\frac{y}{x} \right)^2 \right].$$

Setting $y = vx$, we have

$$x \frac{dv}{dx} + v = \frac{1}{2}(1 + v^2),$$

or

$$2x \frac{dv}{dx} = v^2 - 2v + 1,$$

or

$$\frac{dv}{(v-1)^2} = \frac{1}{2} \frac{1}{x} dx.$$

Then

$$\frac{-1}{v-1} = \frac{1}{2} \ln |x| + c'$$

or

$$v = 1 - \frac{2}{\ln |x| + 2c'}.$$

Replacing v by y/x and setting $c = 2c'$, we have

$$y = x - \frac{2x}{\ln |x| + c}.$$

In determining if a given first-order equation can be written in the form (1.13), the following criterion is often useful. An equation

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (1.14)$$

can be put in the form (1.13) if the functions M and N are *homogeneous of the same degree*. A function $g(x, y)$ (defined for all (x, y)) is said to be *homogeneous of degree m* if

$$g(tx, ty) = t^m g(x, y)$$

for all t .† For example, if

$$g(x, y) = x^3y^2 - 3x^5,$$

we see that

$$g(tx, ty) = t^5(x^3y^2 - 3x^5) = t^5g(x, y).$$

Thus, g is homogeneous of degree 5. However, the function h , where

$$h(x, y) = x^3y - 3x^2,$$

is not homogeneous.

Let us write Eq. (1.14) as

$$\frac{dy}{dx} = -\frac{M(x, y)}{N(x, y)}.$$

If M and N are both homogeneous of degree m , then

$$M(x, y) = t^{-m}M(tx, ty), \quad N(x, y) = t^{-m}N(tx, ty).$$

Setting $t = x^{-1}$, we have

$$M(x, y) = x^m M(1, y/x), \quad N(x, y) = x^m N(1, y/x)$$

and

$$\frac{dy}{dx} = -\frac{M(1, y/x)}{N(1, y/x)}.$$

This equation is of the form (1.13).

In the example

$$x^2 + y^2 - 2x^2 \frac{dy}{dx} = 0,$$

where

$$M(x, y) = x^2 + y^2, \quad N(x, y) = -2x^2,$$

it is clear that M and N are both homogeneous of degree 2. Hence the equation can be written in the form (1.13). Other examples of equations that can be put in the form (1.13) are

$$x^2 \frac{dy}{dx} = y^2 - x^2$$

and

$$(xy + x^2) \sin \frac{y}{x} + y^2 \frac{dy}{dx} = 0.$$

† Some restriction must be placed on the possible values of t if g is not defined everywhere.

Exercises for Section 1.2

In Exercises 1–20, find the general solution, if possible. Otherwise find a relation that defines the solutions implicitly. If an initial condition is specified, also find the particular solution that satisfies the condition.

1. $yy' = 4x$, $y(1) = -3$
2. $xy' = 4y$, $y(1) = -3$
3. $y' = 2xy^2$, $y(2) = 1$
4. $y' = e^x(1 - y^2)^{1/2}$, $y(0) = \frac{1}{2}$
5. $y' = \frac{1 + y^2}{1 + x^2}$, $y(2) = 3$
6. $e^y y' = 4$, $y(0) = 2$
7. $2(y - 1)y' = e^x$, $y(0) = -2$
8. $2y' = y(y - 2)$
9. $3y^2 y' = (1 + y^3) \cos x$
10. $(\cos^2 x)y' = y^2(y - 1) \sin x$
11. $(\cos y)y' = 1$
12. $(\cos^2 x)y' = (1 + y^2)^{1/2}$
13. $x^2 y' = xy - y^2$
14. $x^2 y' = y^2 + 2xy$
15. $xyy' = 2y^2 - x^2$
16. $xy' = y - xe^{y/x}$
17. $e^{y/x} y' = 2(e^{y/x} - 1) + \frac{y}{x} e^{y/x}$
18. $y' = \frac{y}{x} - 3\left(\frac{y}{x}\right)^{4/3}$
19. $xy' = y + (x^2 + y^2)^{1/2}$
20. $3xy^2 y' = 4y^3 - x^3$
21. Show that an equation of the form $y' = F(ay + bx + c)$, $a \neq 0$, becomes separable under the change of dependent variable $v = ay + bx + k$ where k is any number.
22. Use the result of Exercise 21 to solve the differential equation.
 - (a) $y' = (y + 4x - 1)^2$
 - (b) $(y - x + 1)y' = y - x$
 - (c) $(y - 3x)y' = 3(y - 3x + 2)$
 - (d) $(y - 2x)y' = 3y - 6x + 1$
23. Consider a first-order differential equation of the form

$$(a_1 x + b_1 y + c_1)y' = a_2 x + b_2 y + c_2.$$
 - (a) If $a_1 b_2 - b_1 a_2 = 0$, show that the equation is of the type considered in Exercise 21.
 - (b) If $a_1 b_2 - b_1 a_2 \neq 0$ introduce new variables u and v , where

$$u = x + p, \quad v = y + q.$$